



**Kuwait University**  
**Faculty of Science**  
**Department of Mathematics**

**MATH111: Linear Algebra**

**Semester: Summer 2022**

**Midterm 2: Sunday, July 24, 2022, 6:30pm-8pm.**

**Duration: 90 minutes.**

**Solution Key**

| Problem | Points | Score |
|---------|--------|-------|
| 1       | 20     |       |
| 2       | 25     |       |
| 3       | 20     |       |
| 4       | 20     |       |
| 5       | 15     |       |
| Total   | 100    |       |

**Problem 1**

Let  $A^{-1} = \begin{bmatrix} 1 & 0 & 0 & -1 \\ 3 & 1 & 2 & 2 \\ 1 & 0 & -2 & 1 \\ 2 & 0 & 0 & 1 \end{bmatrix}$ .

(a) [5 pts] Find  $|A|$

We use  $\det A = \frac{1}{\det A^{-1}}$ . Regarding  $\det A^{-1}$ , we use REF to make more

zeros:  $\begin{vmatrix} 1 & 0 & 0 & -1 \\ 3 & 1 & 2 & 2 \\ 1 & 0 & -2 & 1 \\ 2 & 0 & 0 & 1 \end{vmatrix} = -3R_1 + R_2 \rightarrow R_2, -R_1 + R_3 \rightarrow R_3, -2R_1 +$

$R_4 \rightarrow R_4, \begin{vmatrix} 1 & 0 & 0 & -1 \\ 0 & 1 & 2 & 5 \\ 0 & 0 & -2 & 2 \\ 0 & 0 & 0 & 3 \end{vmatrix} = (-2)3$  (upper triangular matrix, so product

of the diagonal entries). Therefore  $\det A = \frac{1}{\det A^{-1}} = -\frac{1}{6}$ .

(b) [5 pts] Find  $\text{adj}(A)$

We use the formula  $\frac{1}{\det A} \text{adj}(A) = A^{-1}$ . Multiplying  $\det A$  over yields

$$\text{adj}(A) = \det(A) \cdot A^{-1} = \frac{-1}{6} \cdot \begin{bmatrix} 1 & 0 & 0 & -1 \\ 3 & 1 & 2 & 2 \\ 1 & 0 & -2 & 1 \\ 2 & 0 & 0 & 1 \end{bmatrix}.$$

(c) [5 pts]

Find  $|3A^{-1}A^T|$

We use properties of the determinant such as  $\det(AB) = \det(A) \cdot \det(B)$  and  $\det(A^T) = \det(A)$ . We also use  $\det(\lambda A) = \lambda^n \det(A)$ . So,  $\det(3A^{-1}A^T) = \det(3I_4) \det(A^{-1}) \det(A) = 3^4 = 81$ .

(d) [5 pts] Let  $A$  be an  $n \times n$  matrix. Prove that  $\det(\text{adj}(A)) = (\det(A))^{n-1}$ .

If  $\det(A) \neq 0$  then  $\frac{1}{\det(A)} \text{adj}(A) = A^{-1}$  (book). Rewrite as  $\text{adj}(A) = \det(A)A^{-1}$ . Applying  $\det$  on both sides yields  $\det(\text{adj}(A)) = \det(\det(A) \cdot A^{-1}) = (\det(A))^n \det(A^{-1}) = (\det(A))^n \frac{1}{\det A} = (\det(A))^{n-1}$ . On the other hand, if  $\det(A) = 0$ , then  $\text{adj}(A)$  is singular (because otherwise

the equation  $A \operatorname{adj}(A) = \det(A)I_n$  from the book yields that  $A = 0$  and hence  $\operatorname{adj}(A) = 0$  also, which contradicts the assumption that  $\operatorname{adj}(A)$  is invertible) and hence  $\det(\operatorname{adj}(A)) = 0 = (\det(A))^{n-1}$  also.

## Problem 2

(a) [15 pts] Let  $\mathbf{u} = \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{bmatrix}$  and  $\mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix}$  be vectors in  $\mathbb{R}^n$ , and let  $k \in \mathbb{R}$  be a scalar. Prove:

(i) [5 pts]  $k(\mathbf{u} + \mathbf{v}) = k\mathbf{u} + k\mathbf{v}$

$$\begin{aligned} k(\mathbf{u} + \mathbf{v}) &= k \left( \begin{bmatrix} u_1 \\ \vdots \\ u_n \end{bmatrix} + \begin{bmatrix} v_1 \\ \vdots \\ v_n \end{bmatrix} \right) = k \cdot \begin{bmatrix} u_1 + v_1 \\ \vdots \\ u_n + v_n \end{bmatrix} = \begin{bmatrix} k(u_1 + v_1) \\ \vdots \\ k(u_n + v_n) \end{bmatrix} = \\ &= \begin{bmatrix} ku_1 + kv_1 \\ \vdots \\ ku_n + kv_n \end{bmatrix} = \begin{bmatrix} ku_1 \\ \vdots \\ ku_n \end{bmatrix} + \begin{bmatrix} kv_1 \\ \vdots \\ kv_n \end{bmatrix} = k \begin{bmatrix} u_1 \\ \vdots \\ u_n \end{bmatrix} + k \begin{bmatrix} v_1 \\ \vdots \\ v_n \end{bmatrix} = \\ &= k\mathbf{u} + k\mathbf{v} \end{aligned}$$

(ii) [5 pts]  $\|k\mathbf{u}\| = |k|\|\mathbf{u}\|$

$$\begin{aligned} \|k\mathbf{u}\| &= \sqrt{(ku_1)^2 + \cdots + (ku_n)^2} = \sqrt{k^2((u_1)^2 + \cdots + (u_n)^2)} = |k|\sqrt{(u_1)^2 + \cdots + (u_n)^2} = \\ &= |k|\|\mathbf{u}\| \end{aligned}$$

(iii) [5 pts]  $\frac{1}{\|\mathbf{u}\|}\mathbf{u}$  is a unit vector (assume  $\mathbf{u} \neq \mathbf{0}$ ).

$$\begin{aligned} \text{We check the length of } \frac{1}{\|\mathbf{u}\|}\mathbf{u}, \text{ using the previous property: } \left\| \frac{1}{\|\mathbf{u}\|}\mathbf{u} \right\| &= \\ \left| \frac{1}{\|\mathbf{u}\|} \right| \|\mathbf{u}\| &= 1 \end{aligned}$$

(b) [10 pts]

Let  $V$  be the set of all ordered pairs of real numbers with addition

$$(u_1, u_2) + (v_1, v_2) = (u_1 + v_1, u_2 + v_2 + 1)$$

and scalar multiplication

$$k(u_1, u_2) = (ku_1, ku_2).$$

Check whether  $V$  is a vector space or not. Give reasons for your answer.

We use column vector notation:  $\begin{bmatrix} u_1 \\ u_2 \end{bmatrix} + \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} u_1 + v_1 \\ u_2 + v_2 + 1 \end{bmatrix}$  and

$$k \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = \begin{bmatrix} ku_1 \\ ku_2 \end{bmatrix}$$

So,  $2(\mathbf{u} + \mathbf{v}) = 2\left(\begin{bmatrix} u_1 \\ u_2 \end{bmatrix} + \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}\right) = 2\begin{bmatrix} u_1 + v_1 \\ u_2 + v_2 + 1 \end{bmatrix} = \begin{bmatrix} 2(u_1 + v_1) \\ 2(u_2 + v_2 + 1) \end{bmatrix} =$   
 $\begin{bmatrix} 2u_1 + 2v_1 \\ 2u_2 + 2v_2 + 2 \end{bmatrix} = \begin{bmatrix} 2u_1 + 2v_1 \\ 2u_2 + 2v_2 + 1 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 2u_1 \\ 2u_2 \end{bmatrix} + \begin{bmatrix} 2v_1 \\ 2v_2 \end{bmatrix} +$   
 $\begin{bmatrix} 0 \\ 1 \end{bmatrix} = 2\begin{bmatrix} u_1 \\ u_2 \end{bmatrix} + 2\begin{bmatrix} v_1 \\ v_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} = 2\mathbf{u} + 2\mathbf{v} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} \neq 2\mathbf{u} + 2\mathbf{v}.$  Therefore,  
 it is *not* a vector space.

### Problem 3

Consider the three vectors

$$\mathbf{u} = \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} 1 \\ 4 \\ -2 \end{bmatrix}, \quad \mathbf{w} = \begin{bmatrix} -2 \\ 2 \\ 1 \end{bmatrix}.$$

(a) [10 pts]

(i) [3 pts] Compute  $d(\mathbf{u}, \mathbf{v})$

$$d(\mathbf{u}, \mathbf{v}) = \|\mathbf{u} - \mathbf{v}\| = \left\| \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix} - \begin{bmatrix} 1 \\ 4 \\ -2 \end{bmatrix} \right\| = \left\| \begin{bmatrix} 1 \\ -5 \\ 5 \end{bmatrix} \right\| = \sqrt{1 + 25 + 25} = \sqrt{51}.$$

(ii) [3 pts] Compute  $d(\mathbf{u}, \mathbf{w})$

$$d(\mathbf{u}, \mathbf{w}) = \|\mathbf{u} - \mathbf{w}\| = \left\| \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix} - \begin{bmatrix} -2 \\ 2 \\ 1 \end{bmatrix} \right\| = \left\| \begin{bmatrix} 4 \\ -3 \\ 2 \end{bmatrix} \right\| = \sqrt{16 + 9 + 4} = \sqrt{29}.$$

(iii) [3 pts] Compute  $d(\mathbf{v}, \mathbf{w})$

$$d(\mathbf{v}, \mathbf{w}) = \|\mathbf{v} - \mathbf{w}\| = \left\| \begin{bmatrix} 1 \\ 4 \\ -2 \end{bmatrix} - \begin{bmatrix} -2 \\ 2 \\ 1 \end{bmatrix} \right\| = \left\| \begin{bmatrix} 3 \\ 2 \\ -3 \end{bmatrix} \right\| = \sqrt{9 + 4 + 9} = \sqrt{22}.$$

(iv) [1 pt] Which pair of vectors is closest?

$\mathbf{v}, \mathbf{w}$  is the closest pair because  $\sqrt{22} < \sqrt{29}$  and  $\sqrt{22} < \sqrt{51}$ .

(b) [10 pts] Let  $\mathbf{u} = \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix}$  and  $\mathbf{v} = \begin{bmatrix} 3 \\ -1 \\ -2 \end{bmatrix}$ . Find

(i) [2 pts]  $\mathbf{u} + 2\mathbf{v}$

$$\mathbf{u} + 2\mathbf{v} = \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix} + \begin{bmatrix} 6 \\ -2 \\ -4 \end{bmatrix} = \begin{bmatrix} 7 \\ -3 \\ -2 \end{bmatrix}$$

(ii) [2 pts] A unit vector in the same direction as  $\mathbf{v}$

$$\frac{1}{\|\mathbf{v}\|}\mathbf{v} = \frac{1}{\sqrt{9+1+4}}\mathbf{v} = \frac{1}{\sqrt{14}} \begin{bmatrix} 3 \\ -1 \\ -2 \end{bmatrix}$$

(iii) [2 pts]  $d(\mathbf{u}, \mathbf{v})$

$$d(\mathbf{u}, \mathbf{v}) = \left\| \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix} - \begin{bmatrix} 3 \\ -1 \\ -2 \end{bmatrix} \right\| = \left\| \begin{bmatrix} -2 \\ 0 \\ 4 \end{bmatrix} \right\| = \sqrt{4+16} = \sqrt{20}.$$

(iv) [2 pts]  $\mathbf{u} \bullet \mathbf{v}$ .

$$\mathbf{u} \bullet \mathbf{v} = \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix} \bullet \begin{bmatrix} 3 \\ -1 \\ -2 \end{bmatrix} = 1 \cdot 3 + (-1) \cdot (-1) + 2 \cdot (-2) = 3 + 1 - 4 = 0.$$

(v) [2 pts] the angle  $\theta$  between  $\mathbf{u}$  and  $\mathbf{v}$ .

$$\cos \theta = \frac{\mathbf{u} \bullet \mathbf{v}}{\|\mathbf{u}\|\|\mathbf{v}\|} = 0. \text{ Therefore, } \theta = \frac{\pi}{2}.$$

### Problem 4

(a) [7 pts] Let  $\mathbf{u}$  and  $\mathbf{v}$  be vectors in  $\mathbb{R}^n$ . Show that

$$\|\mathbf{u} + \mathbf{v}\|^2 + \|\mathbf{u} - \mathbf{v}\|^2 = 2(\|\mathbf{u}\|^2 + \|\mathbf{v}\|^2)$$

See book.

(b) [7 pts] Verify the Cauchy-Schwarz inequality for

$$\mathbf{u} = \begin{bmatrix} 1 \\ \frac{1}{\sqrt{3}} \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} 1 \\ -\frac{1}{\sqrt{3}} \end{bmatrix}.$$

Cauchy-Schwarz:  $|\mathbf{u} \bullet \mathbf{v}| \leq \|\mathbf{u}\| \|\mathbf{v}\|$ . Here,  $\mathbf{u} \bullet \mathbf{v} = 1 \cdot 1 + \frac{1}{\sqrt{3}} \cdot \left(-\frac{1}{\sqrt{3}}\right) =$

$1 - \frac{1}{3} = \frac{2}{3}$ . Also,  $\|\mathbf{u}\| = \|\mathbf{v}\| = \sqrt{1^2 + \left(\frac{1}{\sqrt{3}}\right)^2} = \sqrt{1 + \frac{1}{3}} = \sqrt{\frac{4}{3}} = \frac{2}{\sqrt{3}}$ .

Therefore,

$$\frac{2}{3} = |\mathbf{u} \bullet \mathbf{v}| \leq \|\mathbf{u}\| \|\mathbf{v}\| = \frac{2}{\sqrt{3}} \cdot \frac{2}{\sqrt{3}} = \frac{4}{3}$$

which is true.

- (c) [6 pts] Let  $\mathbf{u}$  and  $\mathbf{v}$  be vectors in  $\mathbb{R}^n$ . Under what condition does equality hold in the Cauchy-Schwarz inequality?

The condition is equivalent to  $|\cos \gamma| = 1$  which is if and only if  $\gamma = 0$  or  $\gamma = \pi$ . This is if and only if the vectors are pointing in the same (or opposite) direction.

### Problem 5

[15 pts] Solve for  $x$  in:

$$\begin{vmatrix} x-2 & 2x-3 & 3x-4 \\ x-4 & 2x-9 & 3x-16 \\ x-8 & 2x-27 & 3x-64 \end{vmatrix} = 0$$

Use REF:

$$\begin{vmatrix} x-2 & 2x-3 & 3x-4 \\ x-4 & 2x-9 & 3x-16 \\ x-8 & 2x-27 & 3x-64 \end{vmatrix} = 0 \iff \begin{vmatrix} x-2 & 2x-3 & 3x-4 \\ -2 & -6 & -12 \\ -6 & -24 & -60 \end{vmatrix} = 0 \iff$$

$$(-2)(-6) \begin{vmatrix} x-2 & 2x-3 & 3x-4 \\ 1 & 3 & 6 \\ 1 & 4 & 10 \end{vmatrix} = 0 \iff \begin{vmatrix} x-2 & 2x-3 & 3x-4 \\ 1 & 3 & 6 \\ 0 & 1 & 4 \end{vmatrix} = 0$$

$$\iff (x-2) \begin{vmatrix} 3 & 6 \\ 1 & 4 \end{vmatrix} - 1 \cdot \begin{vmatrix} 2x-3 & 3x-4 \\ 1 & 4 \end{vmatrix} = 0$$

$$\iff (x-2)(12-6) - ((2x-3)4 - (3x-4)) = 0$$

$$\iff 6x - 12 - (8x - 12 - 3x + 4) = 0$$

$$\iff 6x - 12 - 5x + 8 = 0$$

$$\iff x = 4$$